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INFORMATION
TECHNOLOGY

Week 2 - Database Design I

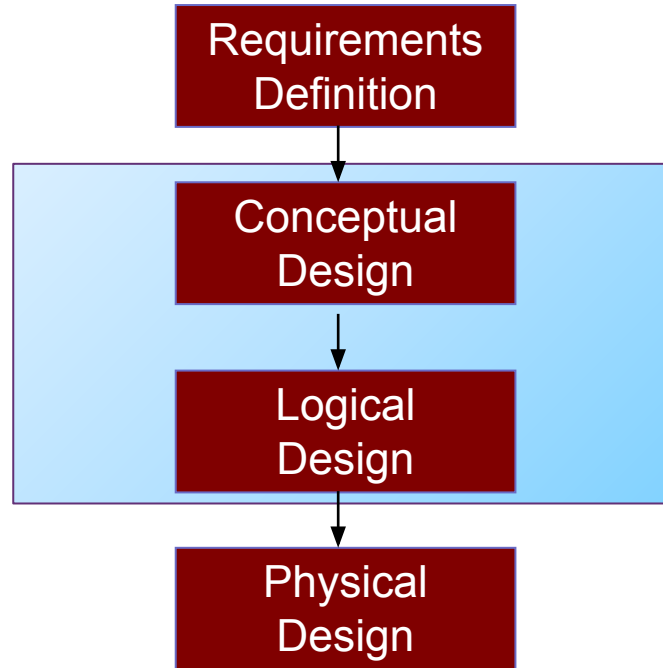
Conceptual Modelling

Workshop 2022 Semester 1

Please obtain a copy of the drone case study for this workshop from the week 2 "Workshop Resources"



The Database Design Life Cycle



Requirements Definition

- Identify and analyse user views.
- A 'user view' may be a report to be produced or a particular type of transaction that should be supported.
- Corresponds to the external level of the ANSI/SPARC architecture.
- Output is a statement of specifications which describes the user views' particular requirements and constraints.

Different views of the underlying data

Web Enrolment System
Enrolment / Re-Enrolment
Fees / Scholarships
Student Services
Course Progression
Enrolment Access Dates
WES Guides
Monash Links
my.monash
Allocate+ (Class Allocation)
Class Timetable (prev. MUTTS)
Moodle

To Add Units Click here

Unit code	Action	Unit name	Campus	Semester	Type	Credits
ACF1200	Change Remove	Accounting for managers PENDING - ENROLLED	CAUL	Semester 1 (2018)	ON-CAMPUS	6
BFF1001	Change Remove	Foundations of finance PENDING - ENROLLED	CAUL	Semester 1 (2018)	ON-CAMPUS	6
BTF1010	Change Remove	Business law PENDING - ENROLLED	CAUL	Semester 1 (2018)	ON-CAMPUS	6
MKF1120	Change Remove	Marketing theory and practice PENDING - ENROLLED	CAUL	Semester 1 (2018)	ON-CAMPUS	6
ECF1100	Change Remove	Microeconomics PENDING - ENROLLED	CAUL	Semester 2 (2018)	ON-CAMPUS	6
ETF1100	Change Remove	Business statistics PENDING - ENROLLED	CAUL	Semester 2 (2018)	ON-CAMPUS	6
MGF1010	Change Remove	Introduction to management PENDING - ENROLLED	CAUL	Semester 2 (2018)	ON-CAMPUS	6
MKF2111	Change Remove	Buyer behaviour PENDING - ENROLLED	CAUL	Semester 2 (2018)	ON-CAMPUS	6
Total credits:						48

If you do not get a **Transaction Number** after you submit, your enrolment is not complete.

SUBMIT ENROLMENT

<https://handbook.monash.edu/2021/units/FIT9132/year=2021>

Overview
Offerings
Rules
Contacts
Learning outcomes
Teaching approach
Assessment
Scheduled teaching activities
Workload requirements
Learning resources

Learning outcomes [Expand all](#)
On successful completion of this unit, you should be able to:

1. Explain the motivations behind the development of database management systems...
2. Describe the underlying theoretical basis of the relational database model and a...
3. Evaluate several design options and construct a database design...
4. Develop a database based on a sound database design...
5. Construct queries that meet user requirements...
6. Contrast the differences between non-relational database models and the relational...

Teaching approach [Expand all](#)
Active learning

Assessment [Expand all](#)
Workshop Exercises
Assignment 1A: Conceptual Model
Assignment 1B: Normalisation and Logical Database Design

Student

Allocate+
Subject Administrator
FIT9132_CL_S2_ON-CAMPUS, INTRO TO DATABASES

Activity Groups:
Applied_PASS-Optional
(Allocation Adjustment)
Enrolments: 240
Preferences: 0
Allocations: 139
Seats Provided: 180
Warning: Not enough seats provided

Tutorial
(Allocation Adjustment)
Enrolments: 240
Preferences: 132
Allocations: 236
Seats Provided: 256

Workshop
(Allocation Adjustment)
Enrolments: 240
Preferences: 133
Allocations: 234
Seats Provided: 280

Show Subject | Special Consideration | Update Details | Section Insert

FIT9132_CL_S2_ON-CAMPUS:INTRO TO DATABASES
Tutorial
Allocate Add Activity Show Message Show Allocated Waitlist Show Unallocated

Functions	Activity Code	Campus	Day	Start Time	End Time	Location	Staff	Duration	Planned Size
Delete Edit List Constraint Context Email	01_OnlineRealTime CL		Tue	18:00	20:00	Online, Melbourne Time	-	120	30
Delete Edit List Constraint Context Email	02_OnCampus CL		Thu	10:00	12:00	CL_Exh-20.Woodside_104	-	120	60
Delete Edit List Constraint Context Email	03_OnCampus CL		Thu	12:00	14:00	CL_Exh-20.Woodside_G06	-	120	60
Delete Edit List Constraint Context Email	06_OnlineRealTime CL		Wed	14:00	16:00	Online, Melbourne Time	-	120	30
Delete Edit List Constraint Context Email	07_OnlineRealTime CL		Tue	14:00	16:00	Online, Melbourne Time	-	120	30
Delete Edit List Constraint Context Email	09_OnlineRealTime CL		Thu	18:00	20:00	Online, Melbourne Time	-	120	30
Delete Edit List Constraint Context Email	10_OnlineRealTime CL		Thu	18:00	20:00	Online, Melbourne Time	-	120	30

Staff & Student

Admin

ER Modeling

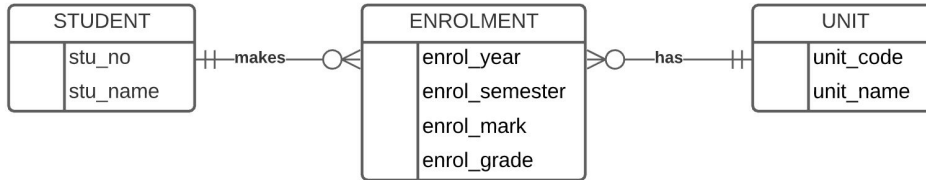
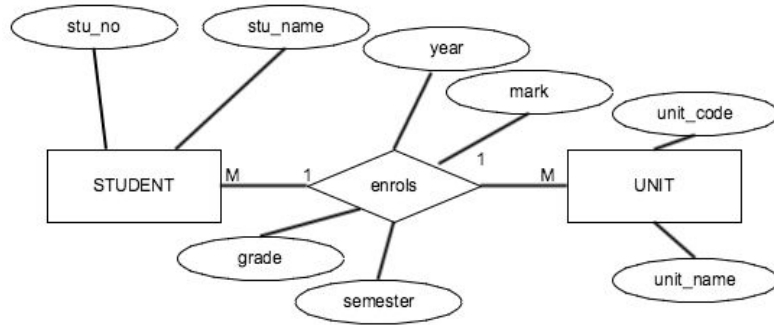
- ER (Entity-Relationship) model developed by Peter Chen in 1976 to aid database design.
- Used for conceptual model (ERD).
- ER diagrams give a visual indication of the design.
- Basic components:
 - Entity
 - Attribute
 - Relationship



Conceptual Design

- Develop the enterprise data model.
- *Independent of all physical implementation considerations (the type of database to be used).*
- In development try to minimise redundancy within provided scenario however must not add anything outside scenario
- Various design methodologies may be employed such as UML, ER (Entity-Relationship) Modelling and Semantic Modelling.
- ER consists of ENTITIES and RELATIONSHIPS between entities
 - An ENTITY will have attributes (things we wish to record), one or more of which will identify an entity instance (called the KEY)

ERD - Notation



Chen

Information
Engineering/James
Martin/Crows foot
**** This is what we will
be using***

Conceptual Level (ER Model)

ENTITY

Collection of "Customer(s)"

KEY ATTRIBUTE(S)

Instance identifier

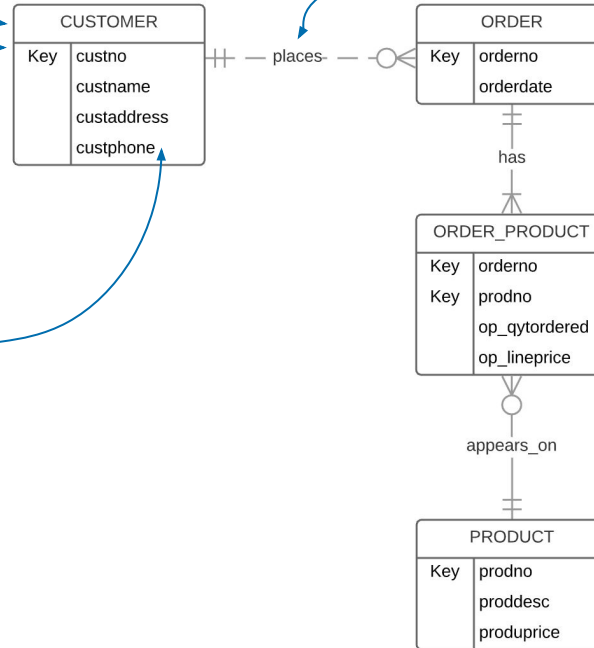
NON KEY ATTRIBUTE

Other non-key attributes

RELATIONSHIP

*Connects entities - on a conceptual model this is the **ONLY** manner in which entities are connected*

Assignment 1A



Q1. In your group discuss your pre workshop identification of the Monash Software Entities and arrive at a group consensus. How many entities exist?

Case study: <https://lms.monash.edu/mod/resource/view.php?id=10073912>

- A. 2
- B. 3
- C. 4
- D. 5
- E. 6

Conceptual Level (Monash Software Entities)

TRAINING	
Key	training_code

TEAM	
Key	team_no

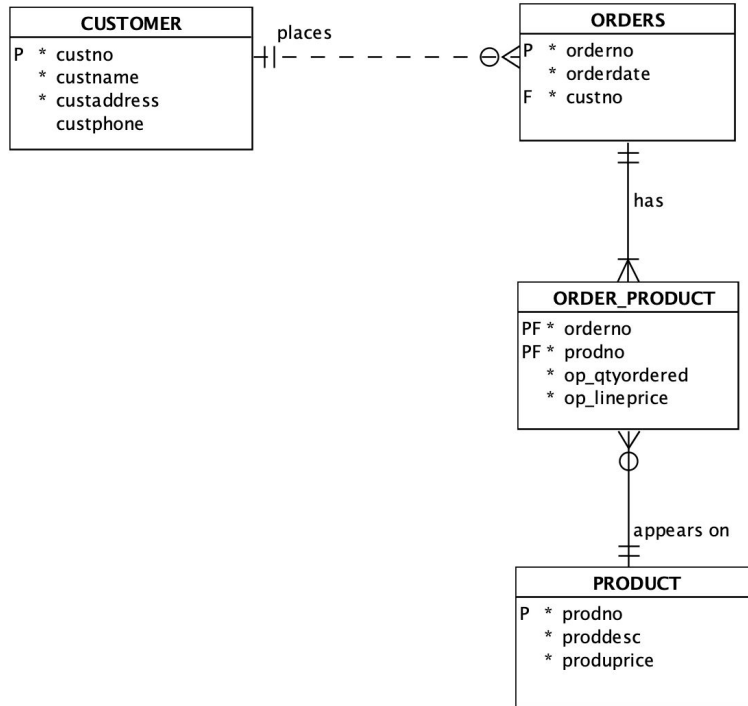
EMPLOYEE	
Key	emp_no

FAMILY_MEMBER	
Key	

Logical Design

- Develop a data model which targets a particular database type (e.g. relational, hierarchical, network, object-oriented, noSQL).
- Independent of any implementation details which are specific to any particular vendors DBMS package.
- Normalisation technique (see week 4) is used to test the correctness of a relational logical model.

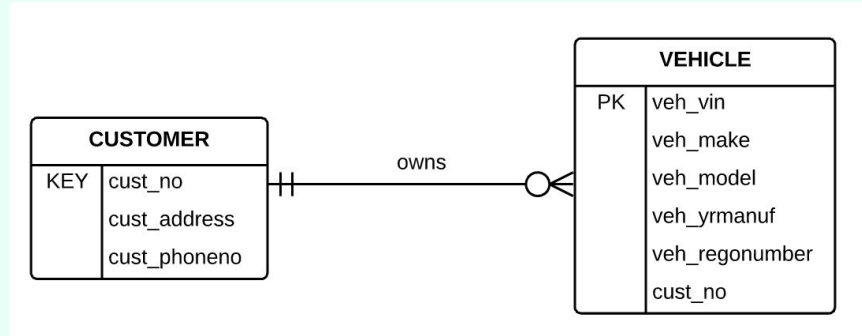
Logical Level (Logical Model - Relational)



Assignment 1B

Q2. Is the diagram shown below a valid *Conceptual Model*?

Be prepared to justify your answer with why you chose this option



- A. Yes
- B. No
- C. Depends on how it is implemented in the database

Physical Design

- Develop a strategy for the physical implementation of the logical data model.
- Choose appropriate storage structures, indexes, file organisations and access methods which will most efficiently support the user requirements (not part of unit).
- Physical design phase is dependent on the particular DBMS in use.

Physical Level – Starting point

```
Oracle Database 12c  Relational_1  Generate

8 CREATE TABLE customer (
9     custno      NUMBER(7) NOT NULL,
10    custname     VARCHAR2(50) NOT NULL,
11    custaddress  VARCHAR2(50) NOT NULL,
12    custphone    CHAR(10)
13 );
14
15 COMMENT ON COLUMN customer.custno IS
16     'Customer number';
17
18 COMMENT ON COLUMN customer.custname IS
19     'Customer name';
20
21 COMMENT ON COLUMN customer.custaddress IS
22     'Customer address';
23
24 COMMENT ON COLUMN customer.custphone IS
25     'Customer phone number';
26
27 ALTER TABLE customer ADD CONSTRAINT customer_pk PRIMARY KEY ( custno );
28
29 CREATE TABLE order_product (
30     orderno      NUMBER(7) NOT NULL,
31     prodno       NUMBER(7) NOT NULL,
32     op_qtyordered NUMBER(3) NOT NULL,
33     op_lineprice  NUMBER(8, 2) NOT NULL
34 );
35
```

The database *schema*

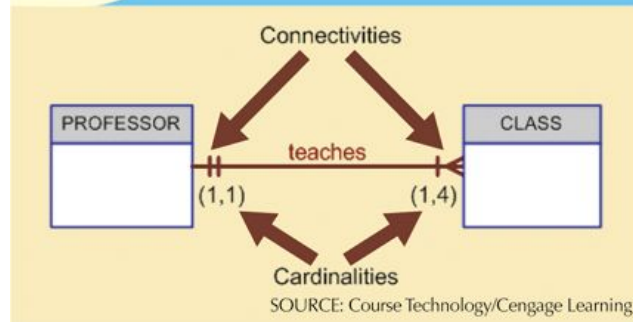
Important rule for Conceptual Modelling

- ***All that is described in the brief has been included and all that has been included was described in the brief***
 - Every entity, attribute and relationship described in the brief **has been included**, and
 - **Must not add** entities, attributes and relationships which are not included as part of the brief, and
- In a real life scenario if there are concerns about features of the brief, discuss with client
 - For assignments:
 - your client will be the ed forum
 - may make assumptions provided they do not violate this rule

CONNECTIVITY/CARDINALITY

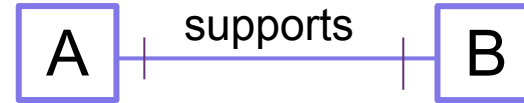
FIGURE 4.7

Connectivity and cardinality in an ERD

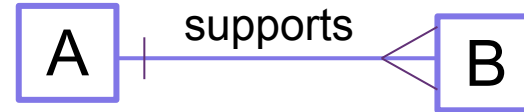


CONNECTIVITY

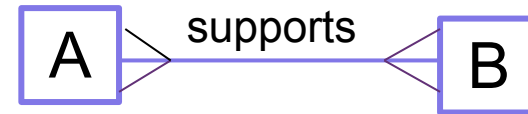
one to one



one to many



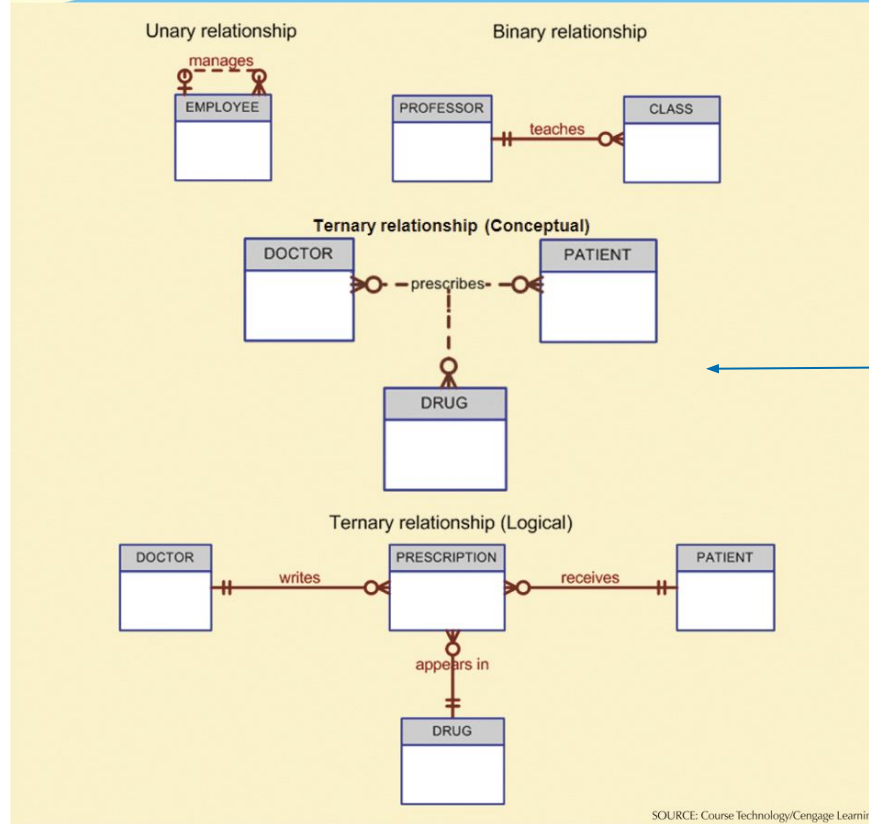
many to many



In general for Crows Foot notation specific cardinalities are not shown as above eg. (1,4), instead cardinality is depicted via min and max using standard symbols (Inside symbol = min, outside symbol = max)

FIGURE
4.15

Three types of relationship degree



Note this is not an acceptable form of a conceptual model in Crow's Foot notation (*relationship lines must not join*)

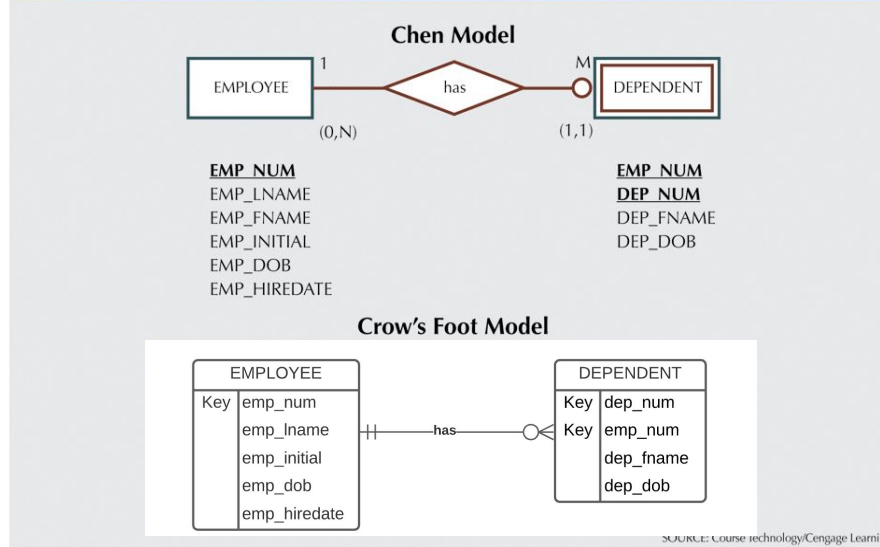
Weak vs Strong Entity

- Strong entity
 - Has a key which may be defined without reference to other entities.
 - For example EMPLOYEE entity.
- Weak entity
 - Has a key which requires the existence of one or more other entities.
 - For example FAMILY entity - need to include the key of employee to create a suitable key for family
- Database designer often determines whether an entity can be described as weak based on business rules
 - customer pays monthly account
 - Key: cust_no, date_paid, or
 - Invented Key: payment_no (surrogate – not at conceptual level)

Weak vs Strong Entity

FIGURE 4.10

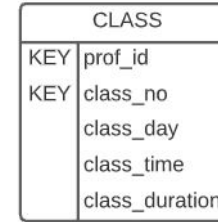
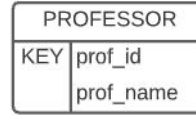
A weak entity in an ERD



Note the Crow's Foot model shown here has been modified from the text version

Q3.The client indicates that a **CLASS** is identified by a combination of the the **prof_id** and the assigned class number for the professor (1st class, 2nd class, 3rd class etc):

prof_id,	class_no,	class_day, ...
1,	1,	Tue
1,	2,	Tue
1,	3,	Wed
2,	1,	Thu
2,	2,	Tue
....		



This business rule is captured in the provided diagram. Pick the correct statement for this diagram.

- A. Both entities are strong entities
- B. PROFESSOR is a strong entity, CLASS is a weak entity
- C. CLASS is a strong entity, PROFESSOR is a weak entity
- D. Both entities are weak entities

Identifying vs Non-Identifying Relationship

- **Identifying**

- Identifier of A is part of identifier of B.



- Shown with solid line
- ENROLMENT - STUDENT
Enrolment key includes student id,
which is an identifier of student.

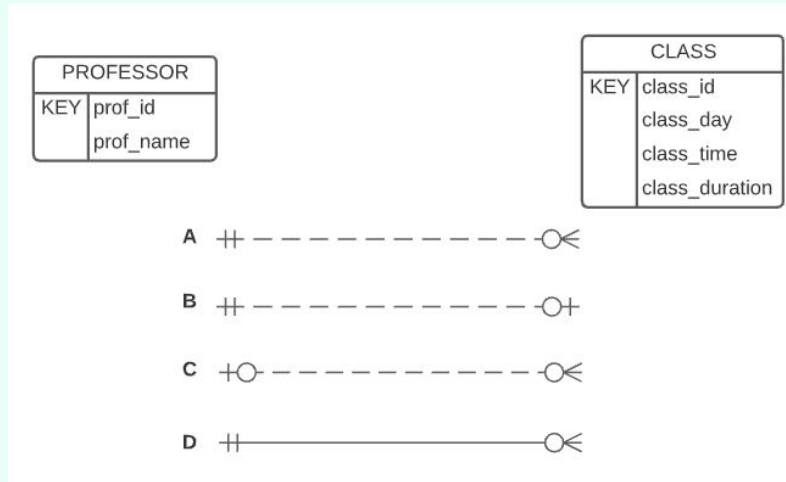
- **Non-identifying**

- Identifier of A is NOT part of identifier of B.



- Shown with broken line
- Department no (identifier of department) is not part of Employee's identifier.

Q4. The client indicates that a professor may teach several classes, but some professors do not have any assigned classes. Each class is taken by only one professor. Note that in this diagram, each class has a unique class id (class_id). Pick the most appropriate relationship for this business rule.



Types of Attributes

- Simple
 - Cannot be subdivided
 - Age, sex, marital status
 - Composite
 - Can be subdivided into additional attributes
 - Address into street, city, zip
 - Single-valued
 - Can have only a single value
 - Person has one social security number
- Multi-valued
 - Can have many values
 - Person may have several college degrees
 - Derived
 - Can be derived with algorithm
 - Age can be derived from date of birth
 - Attribute classification is driven by Client requirements
 - Phone Number?

Q5. The HiFlying case study indicates "*HiFlying establishes a drone hire rate as a cost per hour for customers to rent this particular drone (rates per hour are often changed over the life of the drone, as it ages, although they are only interested in recording the current cost per hour for the drone).*"

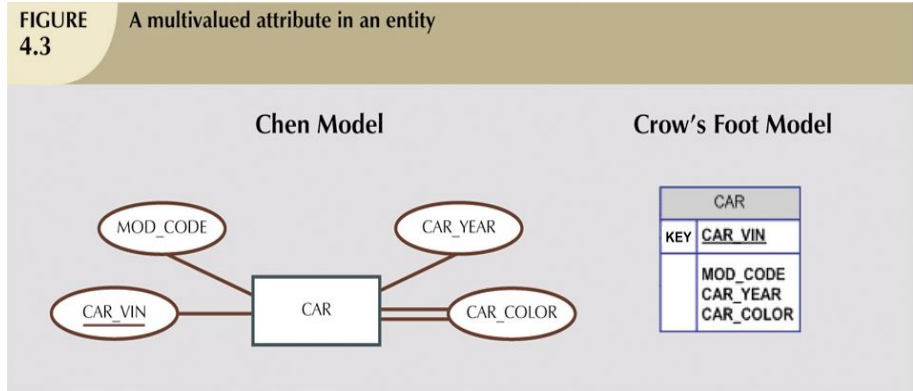
Note that although the hire rate may change over the life of the drone, it is not directly related to the hours flown.

What type of attribute is the drone hire rate?

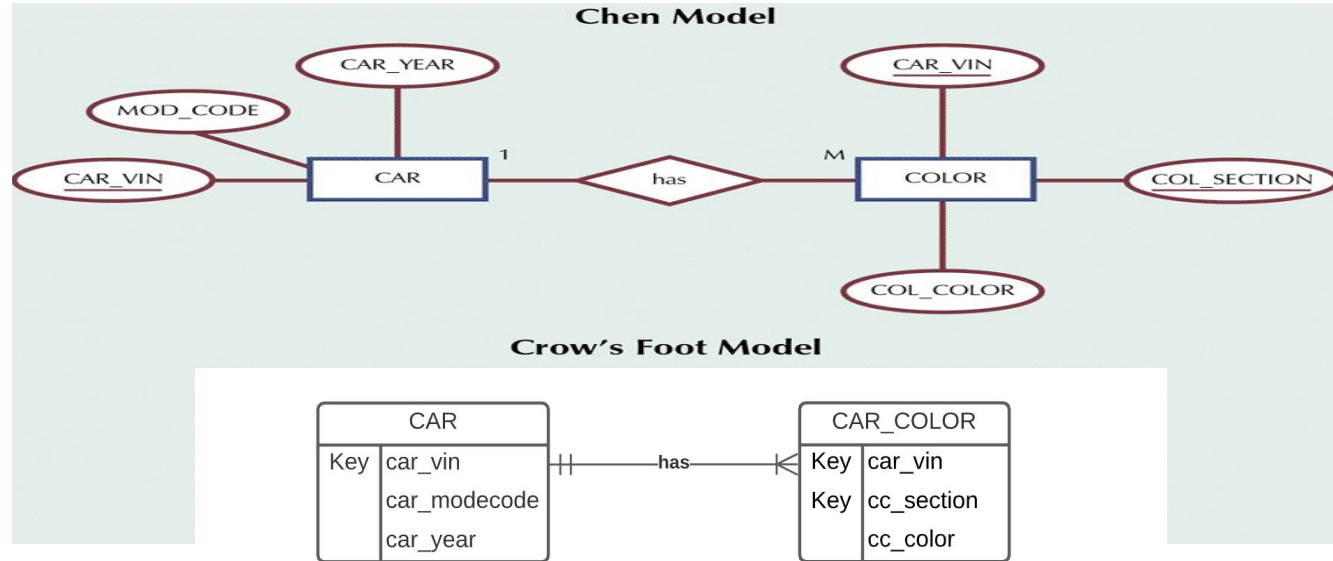
- A. Simple
- B. Composite
- C. Single-valued
- D. Multi-valued
- E. Derived

Multivalued Attribute

- An attribute that has a list of values.
- For example:
 - Car colour may consist of body colour, trim colour, bumper colour.
- Crow's foot notation does not support multivalued attributes. Values are listed as a separate attribute.

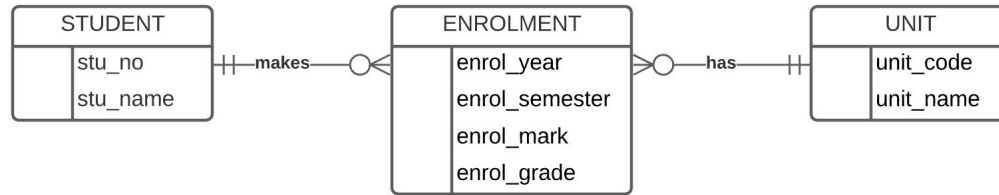
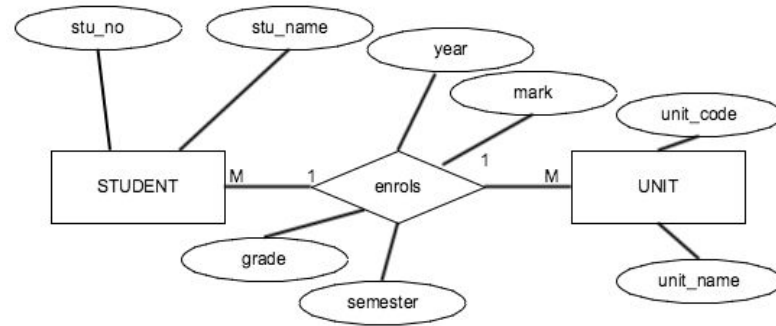


Resolving Multivalued Attributes

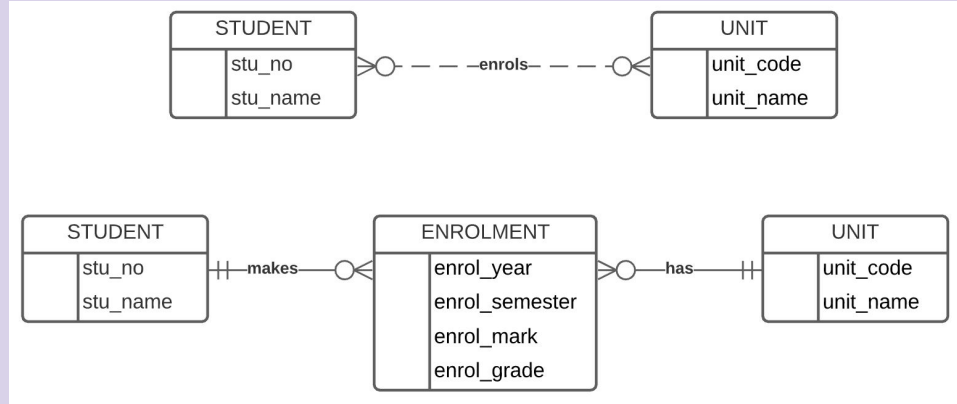


Note the Crow's Foot model shown here has been modified from the text version

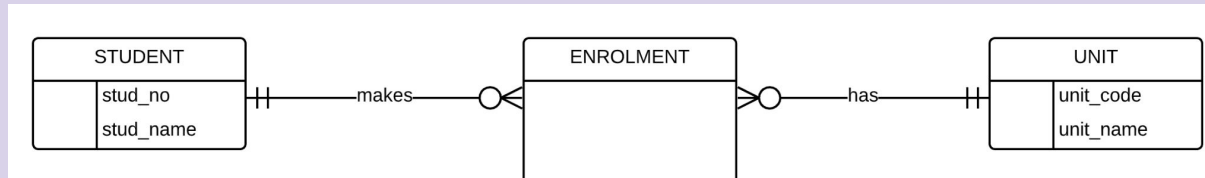
Associative (or Composite) Entity



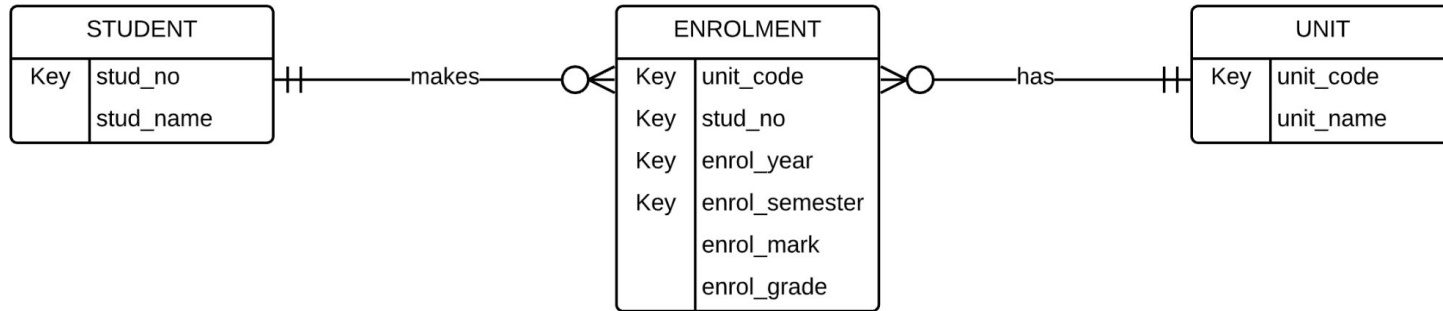
Associative (or Composite) Entity



Q6. Show all attributes for the three entities and add KEYS:



Associative or Composite Entities



Q7. STEP 1: List ALL entities and their key attribute/s which exist in the DRONE case study:

<https://lms.monash.edu/mod/resource/view.php?id=10031949>

For example:

DRONE	
KEY	drone_id

HiFlying Drones



**HiFlying
Drones**

HiFlying Drones is a company which rents drones out to customers.

The company purchases a range of different types of **drones** in order to meet their customers' requirements. Each type of drone they purchase is assigned a drone type code (e.g. PH4) as the identifier for this type.

To keep track of the drones they purchase, HiFlying identifies each drone with a **drone id**. When a new drone is added to the system the type of the drone, the date it was purchased and the purchase price are recorded. In addition, HiFlying establishes a cost per hour for each drone. This cost per hour is later changed over the life of the drone.

HiFlying Drones - Step 1 Identify Main Entities

DRONETYPE	
KEY	dt_code

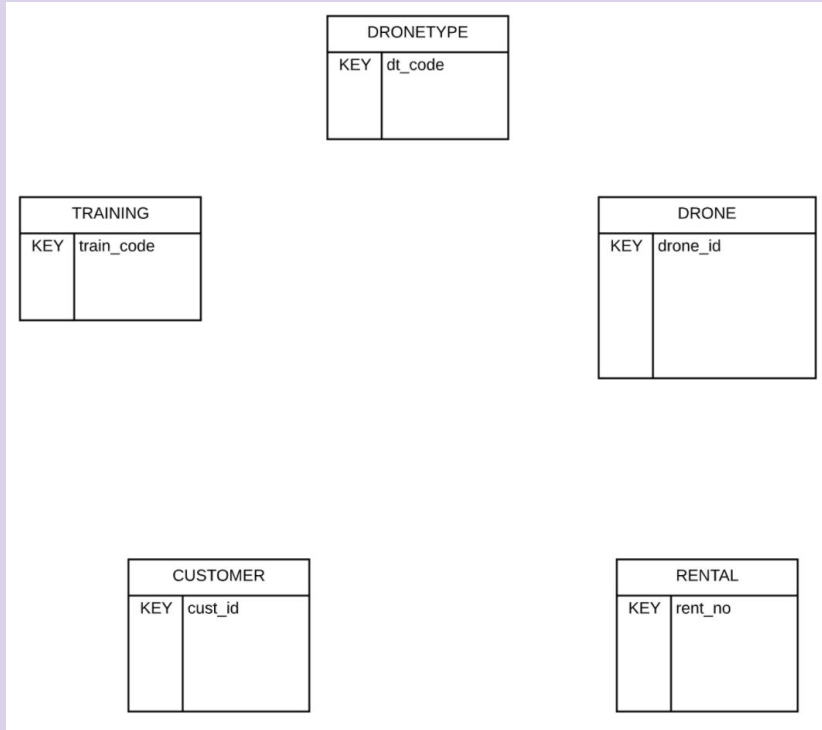
TRAINING	
KEY	train_code

DRONE	
KEY	drone_id

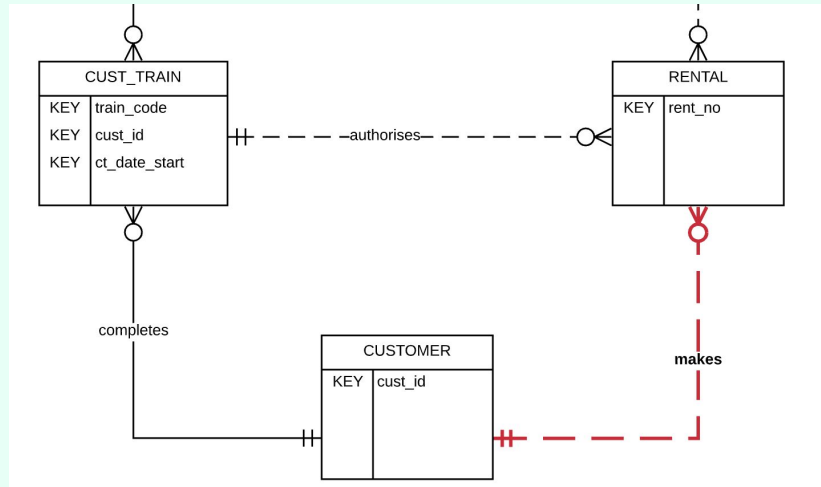
CUSTOMER	
KEY	cust_id

RENTAL	
KEY	rent_no

Q8. STEP 2: Identify the relationships which exist between these entities (remember to add an appropriate verb):

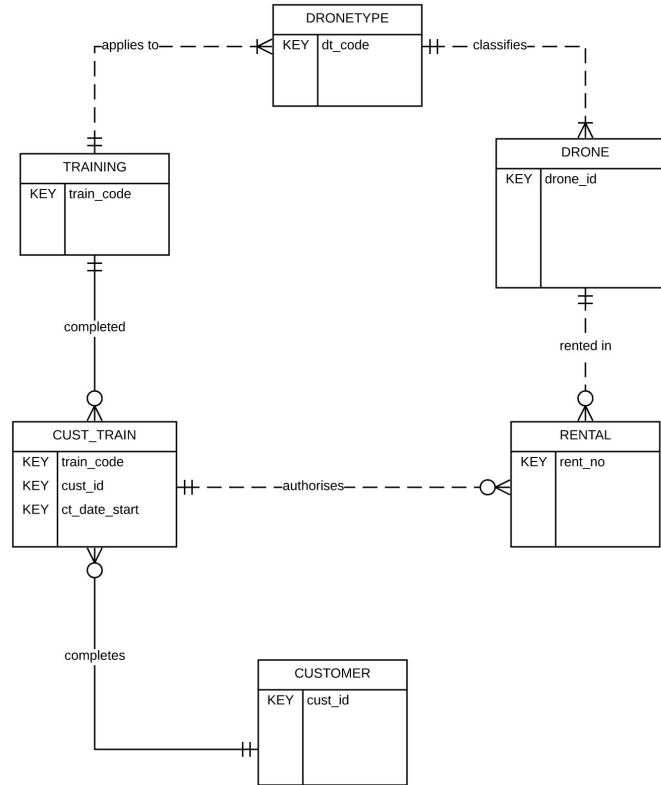


Q9. Since a customer makes a rental, should the database designer include a relationship between RENTAL and CUSTOMER?

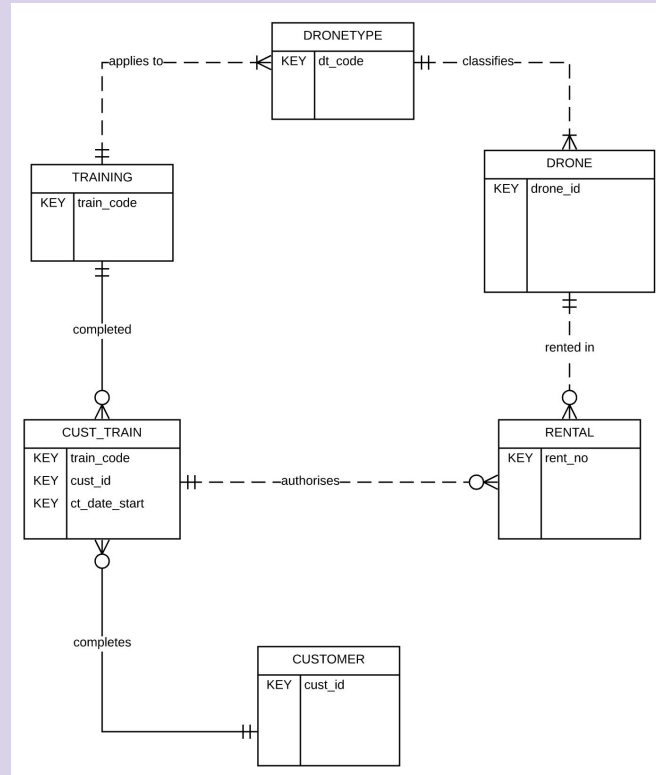


- A. Yes, it is an important relationship to capture
- B. No, it is redundant information
- C. It depends on the client's requirements

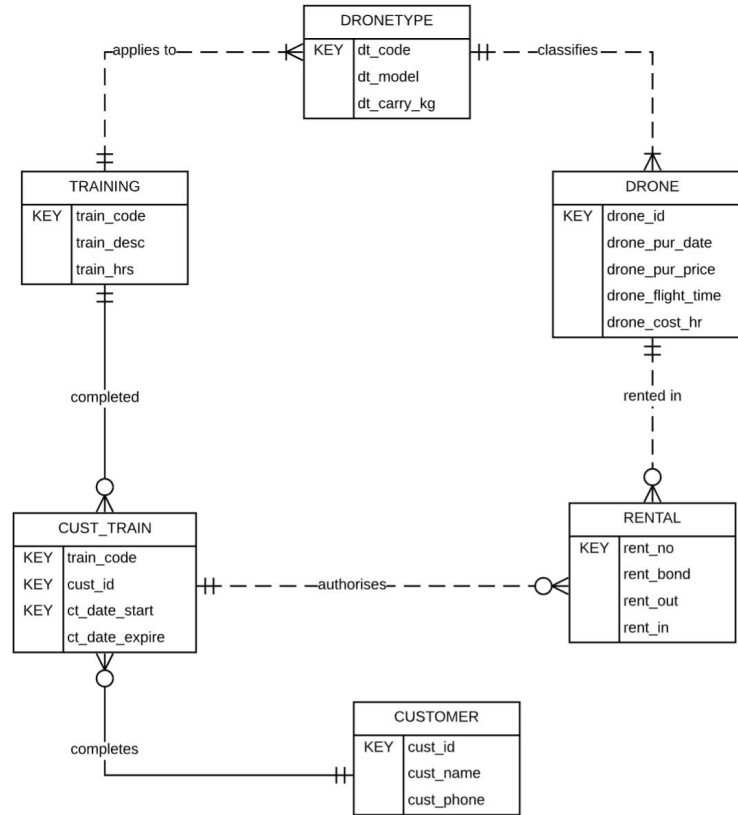
HiFlying Drones - Step 2 Identify Relationships



Q10. Step 3 Add Non-Key Attributes



HiFlying Drones - Step 3 Add Non-Key Attributes - Final Model



Conceptual Model (Monash Software)

You have completed

- **Step 1 identify entities and keys** of the modelling process for Monash Software

TRAINING	
Key	training_code

TEAM	
Key	team_no

EMPLOYEE	
Key	emp_no

After the workshop please proceed and complete:

- **Step 2 Identify Relationships**, and
- **Step 3 Add all non key attributes**

FAMILY_MEMBER	
Key	

A video will be provided showing the full process at the start of next week.